Cognitive and Knowledge Services Architecture Document

Integrated Claims AI Platform · Gate deliverable for the AI platform investment decision

Project	Cognitive and Knowledge Services Architecture
Scenario	Integrated insurance claims AI platform
Learner Name	[Enter name]
Submission Format	Word (.docx) and PDF; diagram embedded as Figure 1 and attached as a vector PDF
Written Content Target	800-1,200 words, not counting the diagram

Assumptions. US multi-state P&C carrier licensed in New York, so 23 NYCRR 500 (NYDFS cybersecurity) and 11 NYCRR 243 (Regulation 152, records retention) are the compliance citations; Data Privacy confirms state variations. All resources run in US Azure regions behind private endpoints, with Entra ID authentication and customer-managed keys.

1. Solution Overview

Overview Paragraph
A large property and casualty insurer is building an integrated Azure AI platform to shorten the time from first notice of loss (FNOL) to adjuster decision, remove manual re-keying errors and give adjusters fast access to policy and regulatory knowledge. It delivers five capabilities: (1) classification and extraction of invoices, structured claims forms, police reports and medical records with Azure AI Document Intelligence; (2) damage-photo analysis for vehicle and property claims with Azure AI Vision, a custom damage detector and GPT-4o-class reasoning; (3) real-time and batch transcription of claims intake calls with Azure AI Speech, plus post-call analytics in Azure AI Language; (4) NLP for customer communications in English, Spanish, French and Portuguese with Azure AI Language and Azure AI Translator; and (5) knowledge mining over a 200,000-document policy and regulatory archive with Azure AI Search, where document, vision and speech outputs also converge into one claims record. The key constraints are regulatory compliance (six-year claim-file retention including call audio, NYDFS access and audit-trail controls, and adjusters seeing only their assigned claims), four-language support for regulated customer communications, and document processing scale, from continuous claim intake to a 200,000-document backfill.

2. Requirements Map

Capability-to-Service Mapping

Capability	Business Requirement	Azure AI Service(s)	Compliance Requirement
Document Processing	Replace manual keying; classify and extract four document types so claims are set up in minutes.	Document Intelligence (classifier; prebuilt, template, neural and composed models); Azure OpenAI fallback; Blob Storage; Functions; AI Search	Medical data and third-party PII restricted and redacted; encryption and least privilege (500.7, 500.15); originals kept as the claim file (243.2(b)(4)).
Vision	Faster, consistent damage triage; early detection of total losses and reused photos.	AI Vision Image Analysis 4.0; custom damage detector (Azure ML AutoML); Azure OpenAI GPT-4o-class vision	AI output is advisory and an adjuster decides (NAIC AI Model Bulletin, 2023); photos with faces, plates or GPS data treated as nonpublic information.

Capability	Business Requirement	Azure AI Service(s)	Compliance Requirement
Speech	Capture FNOL details live; post-call sentiment, entities and summaries; transcribe the call archive.	AI Speech (real-time and batch STT, Custom Speech, language ID); AI Language	**Also applies to NLP and Knowledge Mining:** audio kept 6 years after closure (243.2(b)(4)), then disposed of (500.13); least privilege (500.7); audit trail (500.6); no card codes recorded (PCI DSS 3.3.1).
NLP	Route and prioritise e-mail, chat, SMS and letters in four languages; flag complaints; reply in the customer's language.	AI Language (detection, sentiment, NER, PII, custom classification, CLU); AI Translator (Custom Translator, Document Translation)	PII redacted before storage; complaints feed the complaint register; regulated notices reviewed by a person.
Knowledge Mining	Find, in seconds, the policy wording and regulations that apply to a claim across 200,000 documents.	AI Search (indexers, skillset, hybrid and semantic search, knowledge store); Azure OpenAI embeddings; Functions custom skills	Adjusters see only content for their assigned claims (security filters); every query audit-logged (500.6).

3. NLP Architecture Summary

NLP Design Decisions

Design Area	Learner Response
Azure AI Language capabilities selected	Language detection (routing); PII redaction (before storage); sentiment and opinion mining (escalation, complaint trends); NER (dates, places, vehicles, amounts); key phrases (search tags); multi-label custom text classification for e-mail and letters (new claim, status, documents, payment dispute, coverage, complaint). All run in the source language; the custom model is trained multilingual.
CLU decision	**Yes, for chat, SMS and IVR only.** These short utterances need an intent plus slots (claim number, date of loss) to reach the right queue or self-service API. Intents: CheckClaimStatus, ReportNewClaim, UploadDocument, PaymentQuestion, CoverageQuestion, RequestAdjuster and Complaint. Confidence below 0.70 hands off to an agent. Trained in English with multilingual enabled, so no prior translation. Not used for e-mail, because the 500-character utterance limit and multi-request messages suit custom classification better.
Azure AI Translator integration	Runs **after** native-language analysis, so sentiment and entities come from the customer's own words. **Inbound:** Translator Text adds English beside the original in the claims record. **Attachments:** Document Translation makes English reading copies; extraction uses originals. **Outbound:** English replies are translated with a Custom Translator glossary (deductible → deducible / franchise / franquia); denial letters get bilingual review.
Downstream integration	Message JSON (claimId, language, intents, entities, sentiment, original and English text) goes to the integration layer, then to the claims-record index (artifactType = communication) and Service Bus. From Service Bus it reaches the claims-system queue (complaints go to the complaint register) and Power BI trend reporting.

NLP Architecture Narrative
Messages in four languages reach the right queue without manual reading, and complaints are caught on arrival, which makes handling faster and more consistent. Analysing before translating keeps translation errors out of sentiment and entities. Multilingual training lets one model per task serve all four languages, so Translator is needed only where an adjuster reads or writes.

4. Document Intelligence Architecture Summary

Document Type to Model Mapping

Document Type	Model Selection	Prebuilt / Custom / Composed?	Coverage Boundary Justification	Downstream Field Mapping
Claims forms	Custom template models per form family (auto FNOL, property proof-of-loss, ACORD), composed	Custom template, composed	No prebuilt claims-form model exists, and layout returns untyped key-value pairs. Fixed layouts with checkboxes and signatures suit template models (about 5 samples each; retrain only when a form is revised).	claimNumber, policyNumber, insuredName, dateOfLoss, lossType, lossDescription, estimatedLoss
Invoices	prebuilt-invoice + query fields (ClaimNumber, PolicyNumber)	Prebuilt	Layouts vary but the schema is universal (vendor, dates, line items, totals), and prebuilt covers it in all four languages. Query fields close the insurer-ID gap without training.	vendorName, invoiceId, invoiceDate, invoiceTotal, totalTax, lineItems, claimNumber
Police reports	Custom neural; GPT extraction fallback on layout Markdown	Custom neural	Hundreds of agency layouts mix tables and narrative, and a neural model generalises across them. Unseen formats with key-field confidence below 0.80 fall back to schema-bound GPT extraction, flagged for confirmation.	reportNumber, agency, incidentDate, incidentLocation, parties, vehicles, citations, narrative (PII redacted)
Medical records	Custom neural (header and billing fields) + layout → Text Analytics for health	Custom neural + prebuilt layout	Unstructured, provider-specific narrative with no prebuilt model. The neural model takes the few structured fields; Text Analytics for health extracts diagnoses, procedures and medications with ICD-10 linking.	provider, dateOfService, diagnosisCodes (ICD-10), procedures; sensitivity = medical (injury-claim adjusters only)

Document Classification & Routing Strategy (Custom Classifier / Composed Model)

Custom Document Classifier first; composed model only for claims-form versions. The classifier's splitMode = auto separates multi-document claim packets, and v4.0 composed models cannot include prebuilt-invoice. Classes: invoice, claimsForm, policeReport, medicalRecord, other. An Event Grid BlobCreated event starts a Durable Functions orchestrator that classifies and routes each segment. Invoices go to prebuilt-invoice and claims forms to claims-forms-composed (a new form version is added by re-composing). Police reports and medical records go to their neural models.

Failures: classification below 0.70 or "other" goes to human triage. A key field below 0.80 goes to a validation screen, and the corrections retrain the model. HTTP 429/5xx errors get 5 retries with backoff, then go to a dead-letter queue with an alert. Corrupt files are quarantined and the sender is asked to re-upload.

Document Intelligence Architecture Narrative

Claim set-up moves from manual keying to straight-through extraction, and adjusters review only low-confidence fields, so set-up is faster with fewer errors. Prebuilt is used where a universal schema exists; custom where prebuilt coverage ends. Originals land in a private, CMK-encrypted Blob landing zone (raw-intake, di-results, review-queue, quarantine; malware scan on upload). A normaliser maps the Document Intelligence JSON to the claims-record schema and pushes it to Azure AI Search (mergeOrUpload, artifactType = document). For example, InvoiceTotal.valueCurrency.amount becomes invoiceTotal and DateOfLoss.valueDate becomes dateOfLoss. Each record also carries claimId, docType, sourceUri, page range, confidence and the security fields.

5. Vision Architecture Summary

Vision Design Decisions

Design Area	Learner Response
Azure AI Vision capabilities	Image Analysis 4.0: captions and dense captions (searchable damage descriptions); tags and objects (confirm the photo shows the insured asset); Read OCR (VIN, plates, odometer); multimodal embeddings (find near-duplicate photos across claims); people detection (blur bystanders).
Prebuilt vs custom decision	**Custom for damage; prebuilt for context.** Prebuilt tags say "car" or "roof", not the damage type, location or extent. Required classes: vehicle (dent, scratch, cracked glass, broken lamp, deformation, airbag deployed) and property (shingle loss, hail strikes, water staining, fire and smoke, broken windows, collapse). The custom object-detection model is trained in Azure ML AutoML for Images on adjuster-labelled photos and promoted on per-class recall. Custom Vision was rejected because it retires on 25 Sept 2028.
GPT-4o vision integration decision	**Yes, as an advisory severity layer grounded in detector output.** Severity requires reasoning across several photos and claim context (vehicle age, loss description, estimate), which per-image detection cannot do. GPT-4o-class vision returns JSON: severity (minor / moderate / severe / probable total loss), damaged parts, consistency flags and a rationale citing photo IDs. Temperature is 0, output is labelled "adjuster to confirm", it never denies or settles a claim, and a sample is checked for accuracy each month. GPT-4o versions retired on Azure in 2026, so calls go to a deployment alias, and any successor must pass the same evaluation set.
Output destination	Photo-analysis JSON goes to the claims-record index (artifactType = photo). Severity and the total-loss flag go through Service Bus to the claims system (fast-track; desk or field assignment). Duplicate matches go to the SIU queue, and annotated images go to the adjuster UI.

Vision Architecture Narrative
Photos are triaged minutes after upload on a consistent scale; total losses are fast-tracked and reused photos are flagged before payment. The detector gives accuracy that can be measured per class. GPT-4o-class reasoning adds judgement across all the photos, grounded in that evidence, and an adjuster confirms every grade. Image Analysis 4.0 also retires on 25 Sept 2028, so all Vision calls sit behind one function and the service can be replaced without schema changes.

6. Speech Architecture Summary

Speech Workload Design

Speech Workload	STT Mode	Justification	Downstream Integration
Live claims intake calls	**Real-time** (Speech SDK, diarization, continuous language ID: en-US, es-US, fr-CA, pt-BR)	The value is during the call: live FNOL pre-fill and agent prompts need about one-second latency.	Agent-assist desktop. At call end, the transcript goes to Language analytics, then to the claims record (artifactType = call) and Service Bus.
Historical claims call archive	**Batch** transcription (REST jobs over a blob container, diarization)	No latency need and a large volume. Batch is cheaper, scales by container, writes to company storage and uses the same custom models.	Transcripts container, then Language analytics, then the claims record (claims still in retention) and Power BI QA.

Custom Speech and Compliance Design

Design Area	Learner Response
Custom speech vocabulary	One Custom Speech model per locale, trained on FNOL scripts and claim notes. These cover policy-number and claim-ID formats, coverage terms (deductible, subrogation, actual cash value, PIP, uninsured motorist, endorsement, total loss, salvage), vehicle makes and place names. Training also uses a pronunciation file (ACV, PIP, FNOL, VIN) and 20+ hours of consented 8 kHz call audio. Each call loads a phrase list from the policy record. A model is promoted only if its word error rate beats the base model on a held-out set.
Post-call analytics integration	Transcript goes to Azure AI Language: (1) conversation PII redaction first, so only redacted text is indexed; (2) sentiment per caller turn; (3) NER and key phrases (Text Analytics for health on injury mentions); (4) issue and resolution summary. Insights go to the claims record, to Service Bus (negative sentiment, a complaint or an injury alerts a supervisor) and to Power BI.
Audio retention period	**6 years after claim closure, or until the related market-conduct exam report is filed, whichever is later.** Recordings are claim-file records under 11 NYCRR 243.2(b)(4). A legal hold suspends disposal; at expiry, deletion is secure and logged (23 NYCRR 500.13). For calls with no claim, Privacy sets the period (proposed: 12 months).
Storage location	Dedicated US-region storage account (ZRS, geo-replicated within the US) holding call-audio/{claimId}/{callId}.wav and transcripts/. Private endpoint only, shared keys off, HSM-backed customer-managed keys. Version-level WORM is set to closure + 6 years. Audio moves from Hot to Cool at 30 days and to Cold at 180 days. Speech uses bring-your-own-storage, and endpoint logging is off.
Access controls	No person has standing data access (500.7). Managed identities: Speech can only write, analytics can only read, and the disposal job is the only identity that can delete. Adjusters play audio only through the Claims API, which checks claim assignment and then issues a 15-minute SAS for one blob. QA access is through an Entra group with an ABAC tag condition, and legal access is through PIM with just-in-time approval. Conditional Access enforces MFA (500.12).
Audit logging	Storage, Speech, Language, Key Vault, Search, Entra and PIM logs go to one Log Analytics workspace. The Claims API logs the user, claimId, callId and action, because SAS reads are otherwise anonymous. Logs are kept 6 years (500.6 requires 5) and exported to immutable storage. Sentinel alerts on bulk downloads, access to unassigned claims, and deletes outside the disposal job.

Speech Architecture Narrative
FNOL fields fill in while the caller speaks, and post-call summaries and sentiment focus supervisor review on the calls that need it. Real-time is used only where latency adds value; batch handles the archive more cheaply; both share the same Custom Speech models. Callers hear a recording and AI-transcription notice in their language, card payments move to a DTMF-masked IVR, and only redacted transcripts leave the audio account.

7. Knowledge Mining Architecture Summary

Azure AI Search Indexer Configuration

Configuration Area	Learner Response
Corpus scale	200,000 documents in four virtual-folder partitions (forms and endorsements, guidelines, statutes and bulletins, legacy scans). Each partition has its own indexer, all four share one skillset, so the backfill runs in parallel and faults stay isolated. About 4M clause chunks. An enrichment cache avoids paying again for OCR and embeddings.

Configuration Area	Learner Response
Indexer batch size	**batchSize 20** (blob default 10): with PDFs of about 25 pages, payloads stay within limits while per-batch overhead is halved. Custom skills use a batch of 10, parallelism of 5 and a 90-second timeout. Runs every 5 minutes during backfill, then hourly; runs are time-boxed and resume from the high-water mark.
Change detection	The built-in **LastModified high-water mark** re-processes only new or changed blobs, and **native soft-delete detection** (14-day blob soft delete) removes retired documents. Superseded versions stay indexed with effective and expiration dates, for claims with older dates of loss.
Error handling	**maxFailedItems** is 2,000 (about 1%) for the backfill and 25 in steady state. **maxFailedItemsPerBatch** is 5, so a systemic fault stops the run instead of thousands of documents being skipped. Unsupported or unprocessable files are skipped, and oversized files index metadata only. Failures go to Log Analytics; a nightly job fixes them and re-runs them with Reset Documents.
Access control	**Security filters (GA).** Chunks carry accessScope, policyNumbers, lineOfBusiness and jurisdiction; claims-record artifacts carry claimId and assigned adjuster IDs and groups. Only the Claims API queries Search (managed identity, keys disabled). It maps the adjuster's Entra ID to their assigned claims and adds a mandatory filter. Native document ACLs in Search are still in preview, so they are not used.

Enrichment Skillset Design

Skill	Built-in or Custom?	Business Requirement Supported	Output Field / Destination
Entity recognition	Built-in	Filter by jurisdiction, regulator, date and amount	locations, organizations, dateTimes (facetable); Entities table
Key phrase extraction	Built-in	Coverage concepts as facets and a weighted scoring field	keyPhrases (weight 2); KeyPhrases table
Document classification	Custom (Web API skill → AI Language custom text classification)	Filter by document type and line of business; recency boost for regulations only	docCategory, lineOfBusiness; Documents table
Claims-specific custom skill	Custom (Azure Function: regex + reference data)	Wording in force on the date of loss in the claim's jurisdiction; security-filter fields	formNumber, effectiveDate, expirationDate, jurisdiction, citations, policyNumbers, accessScope
Document Layout	Built-in (Document Intelligence)	OCR legacy scans; chunk by section so answers point to the clause	chunks + sectionHeading (index projections); Clauses table
Azure OpenAI Embedding	Built-in (text-embedding-3-large, 1,024 dims)	Plain-language questions match clauses worded differently	contentVector (HNSW, int8)

Search and Knowledge Store Configuration

Design Area	Learner Response
Hybrid search	BM25 (content, sectionHeading, title, keyPhrases) plus a k = 50 vector query, fused by RRF. This matches adjusters' three query types. Exact identifiers (form "HO 00 03", statute citations) need BM25. Plain-language coverage questions need vectors. Claim-scoped searches add security, jurisdiction, line-of-business and in-force-date filters (vectorFilterMode = preFilter).
Semantic ranking	Reranks the top 50 hybrid results (title = title; content = content, sectionHeading; keywords = keyPhrases) and returns captions and answers that highlight the clause. Used for natural-language queries; identifier lookups skip it.
Scoring profile	**adjusterRelevance:** weights are sectionHeading 3, title 2, keyPhrases 2 and content 1. A tag boost of ×2 applies to the open claim's jurisdiction and lineOfBusiness. A freshness boost of ×1.5 over 365 days applies to regulatoryPublishedDate, which is set only for regulations, so policy forms are not penalised for age. rankingOrder = boostedRerankerScore keeps the boosts after reranking.
Knowledge store projections	**Tables** (Power BI): Documents, Clauses, KeyPhrases and Entities, linked by keys a Shaper skill generates. **Objects:** enriched JSON per document for audit and retraining. **Files:** page images from scans for OCR quality review.

Knowledge Mining Architecture Narrative
Adjusters get the in-force clause, highlighted, in seconds, so coverage decisions are faster and more consistent. Partitioned indexers make the 200,000-document backfill parallel and restartable. The audio access rule from Section 6 is enforced again at query time: transcripts carry the same claimId and adjuster security fields, results never expose storage URLs (playback re-checks assignment), and the Claims API logs every query (user, claimId, document IDs), because Search sees only the API's identity.

8. Multi-Service Integration Diagram

Diagram Instructions
Figure 1 (next page; vector PDF attached) shows the five domains as swim lanes feeding an Azure Functions integration layer. Document, photo, call and communication artifacts converge in the Azure AI Search claims-record index (gold). Downstream, the Claims API serves the Adjuster Search Interface, Service Bus feeds the Claims Management System and Analytics & Reporting, and the skillset feeds the Knowledge Store. Every resource sends logs to Compliance Logging.

Diagram Completeness Check

Required Element	Confirmed?
All five service domains shown	Yes: swim lanes 1–5
Every service has labeled input and output	Yes: each service box has labelled inbound and outbound arrows
Document, vision, and speech outputs converge into Azure AI Search claims record	Yes: document, photo, call and communication artifacts enter the claims-record index
Downstream systems shown	Yes: Adjuster Search Interface, Claims Management System, Analytics & Reporting, Knowledge Store, agent-assist
Compliance logging shown	Yes: dashed audit flows into the Compliance Logging band (Log Analytics, Sentinel, 6-year archive)

Figure 1. Multi-service integration diagram: five service domains, labelled data flows, convergence on the Azure AI Search unified claims record, downstream systems and compliance logging

